User Guide to

Target Manager for Nintendo GameCube™

as used with:

ProDG

Tuner

ProView

SN-TDEV

SN Systems

SN Systems Ltd
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Chapter 1: Introduction

Overview

This user guide describes how to use **Target Manager for Nintendo GameCube**.

Target Manager is used to control access to any Nintendo GameCube development hardware (*targets*) connected to your host PC, or remotely on your LAN. Target Manager must be installed and running in order that other SN Systems applications can communicate with the development hardware.

Target Manager is provided with the following SN Systems products:

- ProDG
- Tuner
- ProView
- SN-TDEV

For more details about any of these products, please contact **sales@snsys.com**.

Technical support and upgrades

First line support for all SN Systems products is provided by the Support areas of our website. To view these pages you must be a registered user with an SN Systems User Name and Password.

- If you have forgotten your User Name and Password, send an e-mail to **webmaster@snsys.com**.

www.snsys.com/support (English)
www.snsys.jp/support (Japanese)

If the answer to your problem cannot be found on the Support areas of our website, you can also e-mail our support team at:

support@snsys.com (English)
j-support@snsys.com (Japanese)

Please make sure that you explain your problem clearly and include details of your software version and hardware setup. If you have been given an SN Systems support log number (LN number) then this should be quoted in all correspondence about the problem.

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Chapter 2: Generic features

Overview of the Target Manager

A *target*, in this context, is Nintendo GameCube development hardware.

- Target Manager allows you to connect to the target, load and run ELF files and view TTY output from the target.
- In order for applications that communicate with the GameCube target to run (ProDG Debugger, ngcrun, etc.) the Target Manager must be running and have at least one target set up.
- Target Manager has been developed so that multiple applications can connect to the target via the same interface, and to allow you to connect to targets, maintain sessions and view session information outside of your particular application.

Why use the Target Manager?

- Target Manager provides a convenient interface for the development of GameCube software, allowing you to run and debug executables, view TTY output from the target and control the DVD emulation drive, all from a single application.
- Target Manager acts as a layer of abstraction between the GameCube hardware and any software that wishes to communicate with the hardware. This allows the Target Manager to trap any dangerous error conditions, and works transparently on all the GameCube development hardware.
- Target Manager provides persistent connection to remote targets on your LAN. If GameCube hardware is scarce, you can set up one PC/GameCube and allow remote users to access the hardware.

Launching the Target Manager

Target Manager can be launched via the command line or via the shortcut in the **Start** menu. It is automatically started when you start ProDG Debugger or Tuner.

The **Start** menu shortcut just accesses the ngctm.exe command line, and can therefore be customized to start the Target Manager in the state that you wish. For example you can specify that the Target Manager always tries to connect to a particular target on start-up.

The Target Manager must be running and have at least one target set up for any of the other tools to be able to communicate with the target GameCube.

ngctm command-line syntax

The Target Manager command line is the following:

```
ngctm [<options> [<args>]]
```

where <options> can be either commands or options that relate to those commands.

The command switches can be any of the following:

Command switches	Actions
-?	Show a list of all the command-line options
-t <target> [<-t args>]	Allows you to modify the parameters of an existing target. For optional -t arguments (<-t args>), see table below.
-m	Starts the Target Manager in a minimized state (system tray icon)
-z <port>	Allows you to specify the port number used by the server to accept remote connections (defaults to 5255).

If the target name contains a space, then <target> must be placed inside quotation marks, e.g. "my ngc".

The command –t <target> is used to indicate a particular target. If you use this command switch, you can follow it with one or more target switch arguments, that enable you to specify what you wish to do to the target.

The -t <target> switch arguments can be any of the following:

-t <target> arguments	**Actions**
-r	Reset the target.
-v	Stop DVD drive emulation.
-x <ELF file> [<-x args>]	Load and run ELF file <ELF file>. For optional arguments, see table below.
-xd <ELF file> [<-xd args>]	Load and debug ELF file <ELF file>. For optional arguments, see table below.
-xr <ELF file> [<-xr args>]	Load and debug ELF file <ELF file>. Unlike –xd, this command will not automatically stop at main.

The –x, -xd and -xr arguments can be any of the following:

-x/xd/xr <ELF file> arguments	**Action**
-a "<ELF args>"	Specify ELF arguments (must be enclosed in quotes).
-d <dvd root>	Set DVD root.
-s "<run args>"	Specify run utility arguments (must be enclosed in quotes). See target manufacturer documents for available arguments.

Target Manager provides command line switches for executing all the necessary command line tools that came with your GameCube hardware. What follows is a summary of how those switches compare with the existing command line tools:

	NPDP-GDEV	**DDH**	**Target Manager**
Run an ELF	odemrun –en <file>	loadrun <file>	ngctm -t <target> -x <file>
Debug an ELF	odemrun –ent <file>	loadrun <file> -T	ngctm -t <target> -xd <file>
Reset	odemsrv –r	reset	ngctm -t <target> -r
Set DVD root	Setodenv <name> <data>	loadrun <file> -r <dvdroot>	ngctm -t <target> -x <file> -d <dvdroot>
Pass ELF arguments	odemrun –en <file> -a <args>	loadrun <file> -a <args>	ngctm -t <target> -x <file> -a <args>
Stop DVD emulation	Odstop	stop	ngctm -t <target> -v

You can pass any additional switches through to the underlying command tool using the Target Manager -s option. All parameters following this switch (which should be enclosed in quotation marks) are passed on to the respective tool. For example, if you want to run an ELF in fast emulation mode on an NPDP-GDEV target, you would pass:

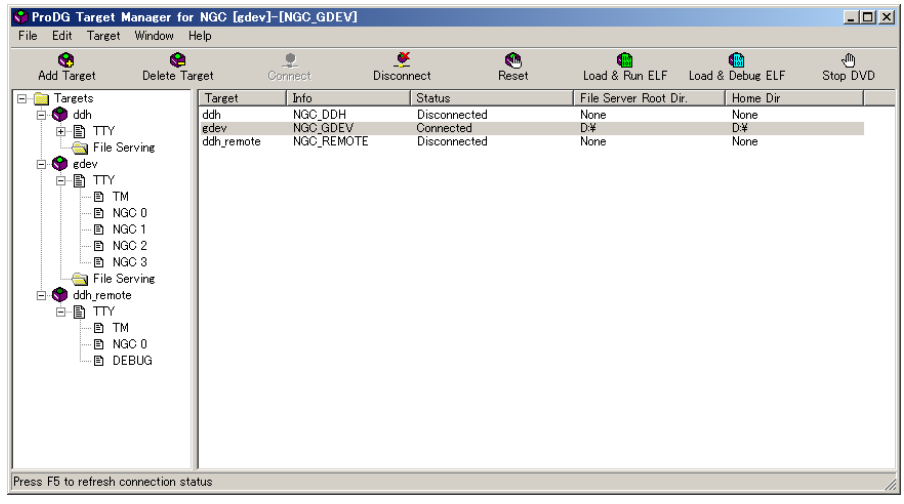
```
ngctm -t <target> -x <file> -s "-f"
```

Can I still use the (NPDP-GDEV/DDH) command line tools?

All communication with the target should go through the Target Manager; this helps the Target Manager to decide when it can safely communicate with the GameCube hardware. So instead of calling the command line tool, you should use the Target Manager equivalent (see the table in the section "ngctm command-line syntax" on page 4).

The Target Manager only ever creates one instance of itself, so repeated calls from the command line will not spawn multiple copies of the Target Manager.

The Target Manager user interface



The Target Manager window includes five menus: **File**, **Edit**, **Target**, **Window** and **Help**. These contain the commands that enable you to add and configure targets, connect to them and load files on selected targets.

Below the menu bar is a toolbar containing buttons that enable you to access the most useful commands rapidly.

The rest of the Target Manager window resembles Windows Explorer and shows any GameCube targets that you might have added on the left and their properties in the right-hand part of the window.

In the left-hand part of the window, the target and its TTY channels are displayed in a tree view.

In the right-hand part of the window, the target properties are arranged in columns labelled **Target**, **Info**, **Status**, **File Server Root Dir.** and **Home Dir**. The column widths may be resized by dragging the column separators either left or right.

To update target connection status

At any time you can refresh display of the current connection status and directory settings for all of the available targets, by pressing the shortcut key <F5>. Pressing <F5> should always result in the connect time for connected targets being updated, and any targets that have been newly connected or disconnected.

To sort rows in Target Manager main window

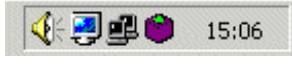
You can sort the rows displayed on the right by clicking on the **Target** column heading. This will sort the listed targets alphabetically by their target name.

To expand out a target

You can view the different TTY output streams of any target, by expanding the target folder in the left part of the window, then expanding the TTY folder, and finally selecting the module whose TTY output you wish to monitor. TTY output will then be sent to the right part of the window.

Target manager tray icon

When you minimize the Target Manager it becomes a system tray icon. If you wish to open the main window again at any time, you can either double-click the tray icon or click the **Open** command in the tray icon shortcut menu:



Keyboard shortcuts

Many of the most frequently used commands in the Target Manager can be accessed via keyboard shortcuts. Currently the keyboard shortcuts are not customizable.

For detailed information on Target Manager keyboard shortcuts, see "Keyboard shortcuts" on page 43.

Exiting the Target Manager

The Target Manager can either be closed from a menu option in the main window or through its tray icon shortcut menu.

To exit the Target Manager

1. If the Target Manager main window is displayed, click **Close** in the **File** menu. However if the Target Manager is minimized you can click **Exit** in the icon shortcut menu.

2. In both cases a dialog appears asking if you are sure that you wish to exit the Target Manager.

3. Click **OK**.

Note: If you are connected to a target when you exit the Target Manager, this connection will be terminated.

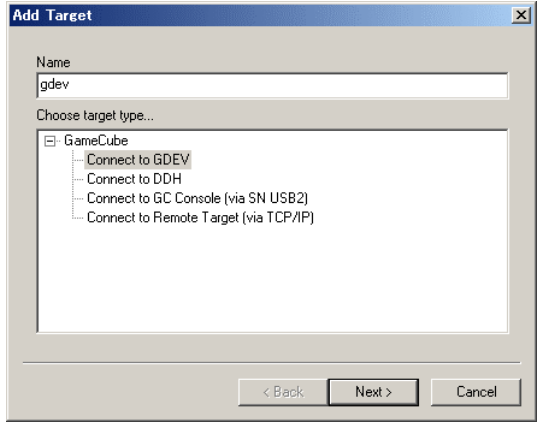
Adding and removing targets

This section describes how you set up the targets in the Target Manager to enable you to connect to the different GameCube targets in your network.

Note: Target Manager uses environment variables to add NPDP-GDEV and DDH targets. You must have successfully installed your Nintendo hardware DDK before you can add targets.

To add a new target

1. Click **Add Target** in the **File** menu (or use the **Add Target** toolbar button, or the <Insert> shortcut key), and the following dialog is displayed:



Available target types are:

- **NPDP-GDEV** - selecting an NPDP-GDEV target will prompt you to choose the COM port that you will use to receive TTY output from the target. The NPDP-GDEV hardware must be connected to your local PC and configured to output TTY to one of your serial ports.

- **DDH** - selecting a DDH target will prompt you to choose the COM port that you will use to receive TTY output from the target. The DDH hardware must be connected to your local PC and configured to output TTY to one of your serial ports.

- **GC console (via SN USB2)** – selecting a USB2 target will prompt you to select from the available USB2 devices connected to your PC. **Either a NPDP or TDEV console must be connected and powered via the supplied USB2 cable.** See "Appendix A1: TDEV console" on page 37 and "Appendix A2: NPDP console" on page 41 for further details.

- **Remote Target (via TCP/IP)** - selecting a remote target type will prompt you to enter the IP address of the server you wish to connect to. The server must be connected to your LAN and be running the Target Manager with at least one available target. Once connected the wizard will display a list of available targets from which you can select.

2. The final wizard page displays a summary of the target you have chosen to add. You can alter these details at any time by clicking the back button. After completing the wizard your new target will appear in the left-hand pane.

Note: It is possible to change the properties of a target when it is not connected. For more information see "Configuring targets" on page 10.

To remove a target

1. Click on the target that you wish to delete in the main window.

2. Click **Delete Target** in the **File** menu (or use the **Delete Target** toolbar button, or use the <Delete> shortcut key). A dialog is displayed asking you to confirm that you wish to delete the target.

3. Click **OK**.

   The target is deleted. If you were connected to the target when you delete the session then you are automatically disconnected.

Configuring targets

If you wish to change any of the properties of an existing target in a single dialog, you can do this in the target property sheet. However, you will not be able to change any connection properties if you are already connected to the target; they are greyed out in the dialog.

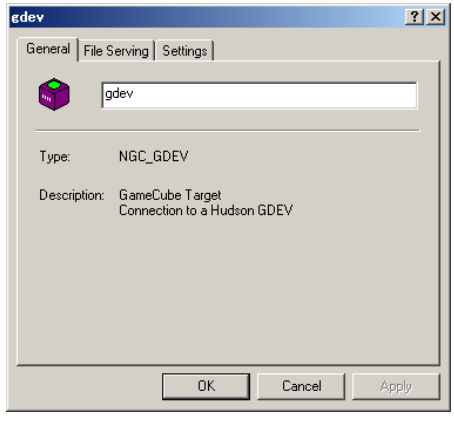
The Target Properties dialog changes according to whether you are viewing properties for local or remote targets, so these are considered separately.

To modify the target properties (local target)

1. Select the required target in the left hand list.

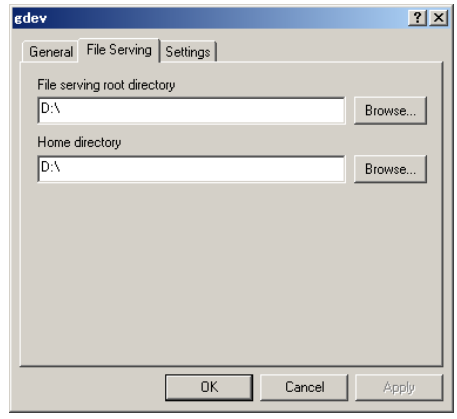
2. Click **Target Properties** in the **Target** menu, or right-click to display the shortcut menu and then select **Properties**, or use the <Alt+enter> shortcut key. The property sheet for the selected target is displayed.

   The following dialog will be displayed. It is divided into three tabs.



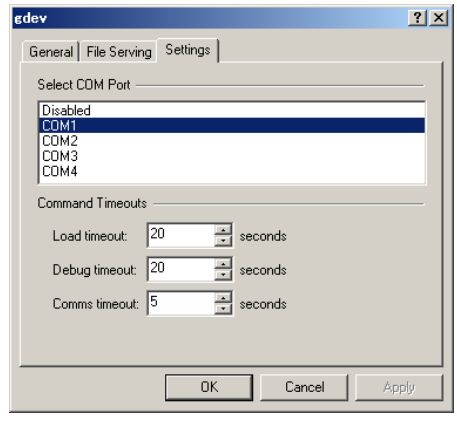
3. The **General** tab displays information about the current target. You can change the target name by typing the new name into the edit box.

4. Select the **File Serving** tab to display the following properties page:



5. This page displays information about file serving settings. You can change the root and home directories by clicking the **Browse** buttons or by entering the values directly into the edit boxes.

6. Select the **Settings** tab to display the following properties page:



7. You can select the COM port that will be used to capture serial I/O from the target from the displayed list.

8. You can also specify the following command timeouts:

Load timeout	This timeout occurs when an ELF file is loaded. You may need to increase this value for large ELFs or projects that contain many files in the dvdroot directory.
Debug timeout	After an ELF has been loaded the Target Manager waits for the debug stub to signal that it is ready. If the stub fails to load then this timeout will occur. You may need to increase this value for projects that contain many files in the dvdroot directory.
Comms timeout	Applies to any command sent from the Target Manager to the debug stub. It will occur if the stub

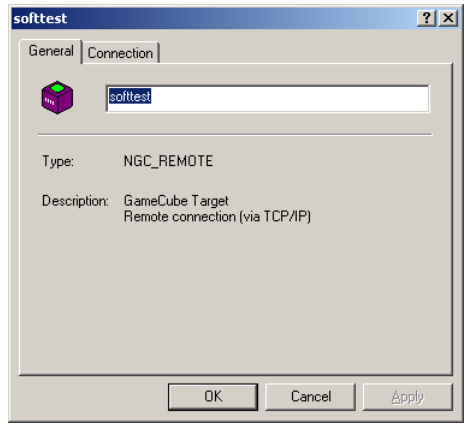
timeout has become corrupted and doesn't acknowledge the command.

To modify the target properties (remote target)

1. Select the required target in the left hand list.

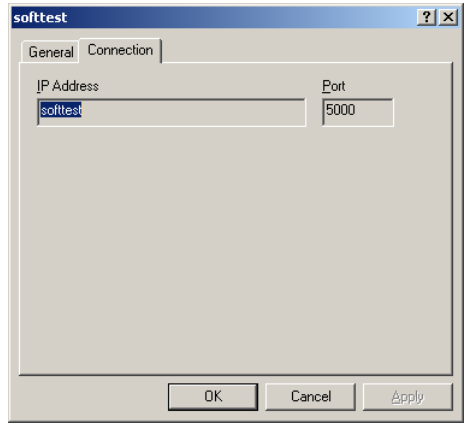
2. Click **Target Properties** in the **Target** menu, or right-click to display the shortcut menu and then select **Properties**, or use the <Alt+enter> shortcut key. The property sheet for the selected target is displayed.

   The following dialog will be displayed. It is divided into two tabs.



3. The **General** tab displays information about the current target. You can change the target name by typing the new name into the edit box.

4. Select the **Connection** tab to display the following properties page:



This page displays information about the connection, i.e. the **IP Address** and **Port** number.

Connecting to targets

Target Manger can either be launched via the command line or Windows shortcut, or it is launched automatically by the ProDG Debugger. For any other tools (Debugger or ngcrun) to be able to interact with a particular target, it must already be configured in the Target Manager.

When the Debugger or ngcrun exit, by default they will leave the target in the state that it was in when they started. So, for example, if you connect to an existing session when you start the Debugger, the session will remain connected when you quit the Debugger. However you can modify this behavior using options on their command lines to specify, for example, that you are always disconnected.

To connect to a target

1. Select the target that you would like to connect to in the right-hand list of the Target Manager main window.

2. Double-click on the target name, or click **Connect** in the **Target** menu (or click the **Connect** toolbar button, or the <C> shortcut key).

   If your connection was successful then the target properties in the right-hand part of the window will be updated to show 'Connected' in the **Status** column. However if another user is currently connected to the target a message is displayed saying that the target is in use by another user, and the identity of the user that is currently connected is displayed in the **Status** column. The **Status** field for another connected user takes the form:

   ```
   Used by [Mike]
   ```

Disconnecting from targets

When the Debugger or the ngcrun command exit, then you may still be connected to the GameCube target in the Target Manager. If you wish to disconnect from the target because you have finished working with it, then you will need to do this in the Target Manager.

To disconnect from a target

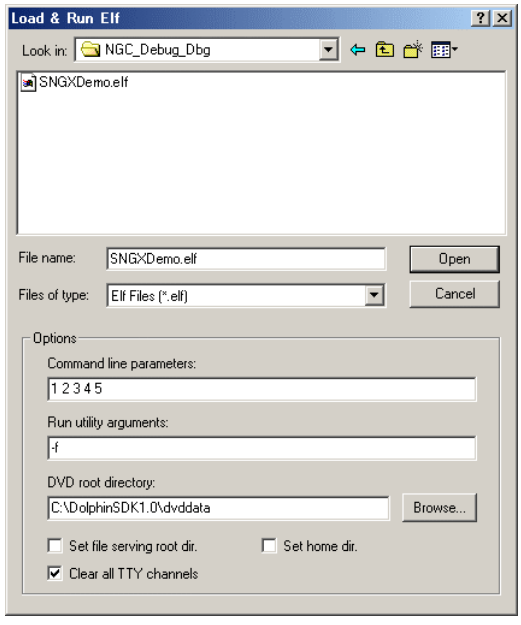
1. Select the connected target that you would like to disconnect from in the right hand list of the Target Manager main window.

2. Double-click on the target name, or click **Disconnect** in the **Target** menu (or click the **Disconnect** toolbar button, or the <d>

shortcut key). The target properties will update to show Disconnected in the **Status** column.

Loading and running ELF files

You can load and run ELF files on the target directly from the Target Manager. To do this click the **Load and Run ELF** button on the Target Manager toolbar, or click the command in the **Target** menu. A dialog appears in which you can select the ELF file to be loaded:



You can also specify the following options:

Command line parameters you can enter a list of arguments to be passed to the ELF file when it is run.

Run utility arguments *(applies only to DDH and NPDP-GDEV targets)* you can pass additional switches to the command line tool that will be used to run the ELF file. For a complete list of the available switches, see the documentation for the tool you are using.

DVD root directory allows you to choose a directory to be used as the DVD emulation drive.

Set file serving root dir. sets the current directory as the file server root directory.

Set home dir. sets the current directory as the home directory.

Clear all TTY channels will clear all TTY channels for the current target before loading the ELF file.

Resetting the target

You can reset the target at any time by clicking the **Reset** button on the toolbar. If the Debugger is currently connected to the target that you are resetting, this will restart the current debugging session.

DVD emulation

During game execution the GameCube hardware will emulate a DVD drive using the dvdroot directory on your PC.

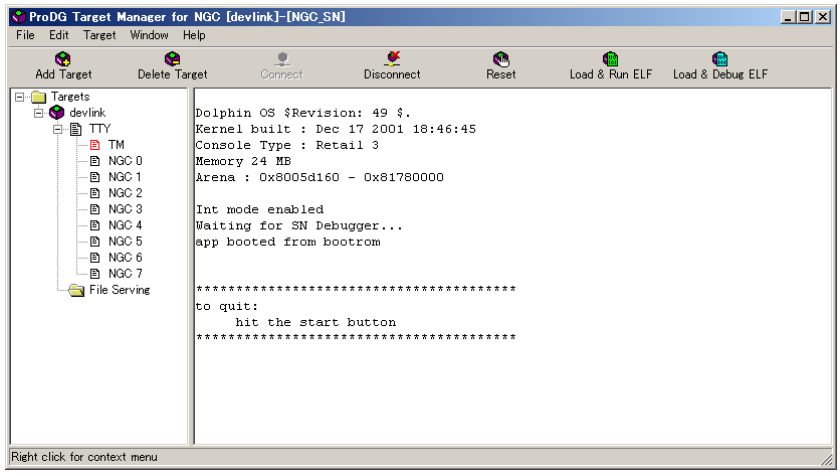
> **Note:** If the target is either NPDP-GDEV or DDH prior to AMC DDK v3.2, any file that is accessed during DVD emulation gets locked and is only unlocked when emulation is stopped or reset. In order to free up access to the files in your dvdroot directory, press the **Stop DVD** toolbar button. Running the ELF file again will cause the dvdroot directory to be locked again.

TTY support

Target Manager provides TTY support via the standard file serving functions. Calling the PCwrite() function with a file handle between -1 and -7 will output text to the corresponding (positive) TTY channel in the Target Manager. The NGC0 channel is reserved for system-generated output.

For example:

```
PCwrite(-1, "Hello Channel 1.\n");
PCwrite(-4, "Hello Channel 4.\n");
```



Support for the file serving functions is provided by the libsn.a library.

Note: Outputting too much TTY from the GameCube target will severely affect the performance of the target. Do not attempt to output TTY from within areas of your program that will be called repeatedly in quick succession.

Viewing output from the GameCube

You can expand each target node listed in the left part of the window to show the content of the different output streams originating from the GameCube processor.

If you are not currently connected to the selected target, the TTY panes will still show any output resulting from a previous connection.

The first level under the target node contains the **TTY** directory.

Selecting the COM Port

Before you can view the TTY output from a local target you must specify the COM port from which you wish to receive the output.

To select the COM Port

1. Select **Target Properties** from the Target menu.
2. Select the **Settings** tab.
3. Select the required COM Port from the displayed dialog.
4. Click **OK**.

Viewing TTY stream output

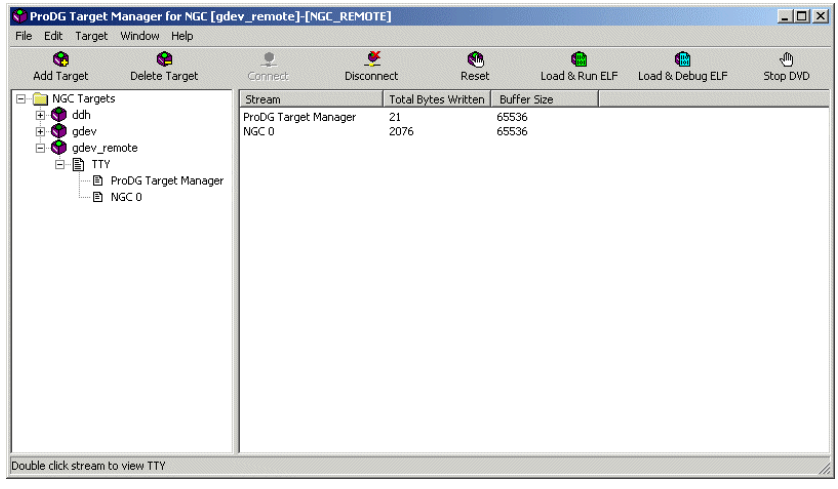
You can view a summary of the TTY channels, or each TTY channel can be displayed individually. You can also display a context menu for a particular TTY channel. This provides further functionality, for example, it enables you to stream the TTY output to a specified file and view channel properties.

To view a summary of TTY channels

Click on the **TTY** directory to select it.

The right-hand pane switches to show a list of the TTY channels available: **Target Manager** and **NGC0 – NGC7**.

For each channel the **Total Bytes Written** column contains the number of output bytes to that channel, while the **Buffer Size** column shows the size of the TTY buffer for the channel:



To view an individual TTY channel

This procedure assumes that you are already viewing the TTY node for a chosen target (see "To view a summary of TTY channels" on page 17).

There are two ways to view an individual TTY channel:

- In the list of TTY channels in the right-hand pane, double-click on a row to select a particular channel.

  or

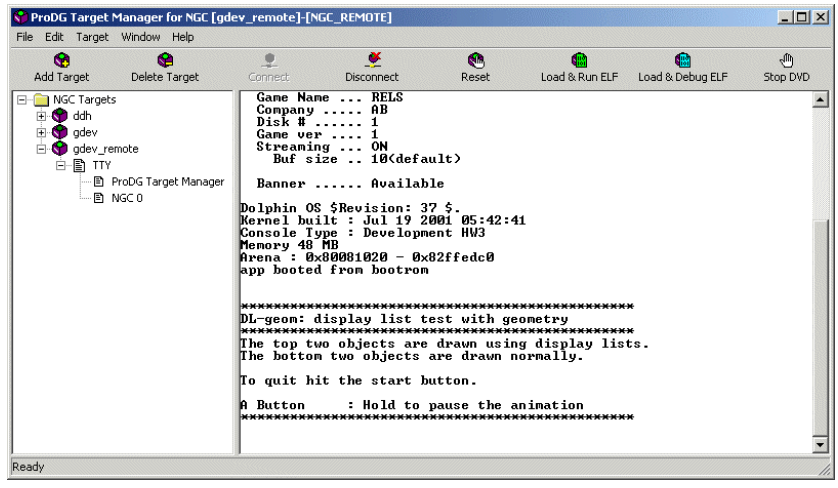
- Double-click on the **TTY** directory, or click on the + sign next to it.

  The **TTY** directory expands to show a tree view of all TTY output channels available.

When a TTY pane contains new output it is indicated by a red icon in the TTY channel list.

Select an individual TTY channel from the list presented.

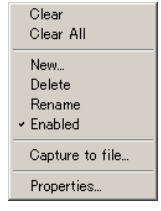
The right-hand pane switches to show any TTY output from that channel:



TTY channel shortcut menu

- Right-click the required TTY channel in the right-hand pane.

  The following shortcut menu will be displayed:



Clear	Clears the currently selected TTY channel.
Clear All	Clears all TTY channels for the selected target.
New...	Creates a new TTY channel.
Delete	Causes the current TTY channel to be permanently deleted.
Rename	Enables you to rename the channel.
Enabled	Toggle to determine whether the TTY channel is active. When disabled, the channel no longer appears in the left-hand tree view and TTY output is not captured for that channel.
Capture to	Allows you to stream TTY output to a file on the

file...	disk. See "To capture to a file" on page 19.
Properties...	Opens the TTY channel's property page. See "To view channel properties" on page 19.

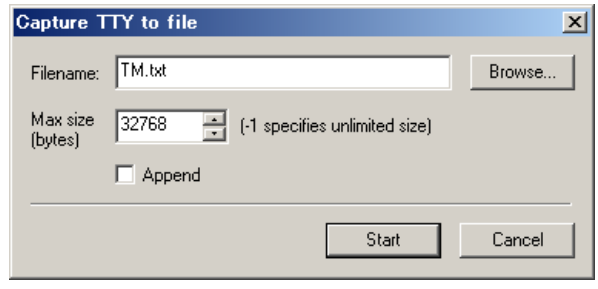
To capture to a file

You can stream TTY output from a particular channel to a specified file as follows:

1. Right-click the TTY channel in the right-hand pane for which you wish to capture TTY output.

2. From the displayed context menu select **Capture to file...**

   The following dialog will be displayed:



3. Select the file that you wish to stream TTY output to.

4. Specify the maximum limit for the size of the file. Enter `-1` for an unlimited size.

5. Check **Append** to place the output after any existing data in the file. If this box is not checked any existing data will be overwritten.

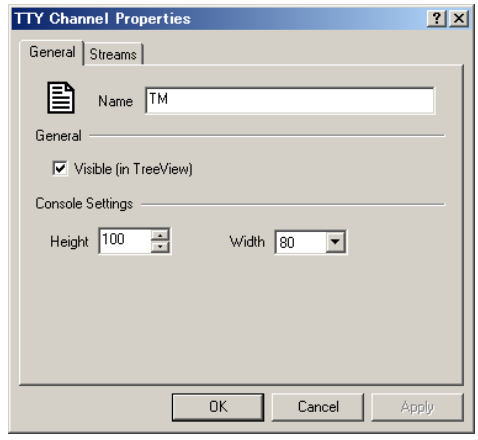
6. Click **Start** to begin the file capture.

To view channel properties

1. Right-click the TTY channel in the right-hand pane for which you wish to view channel properties.

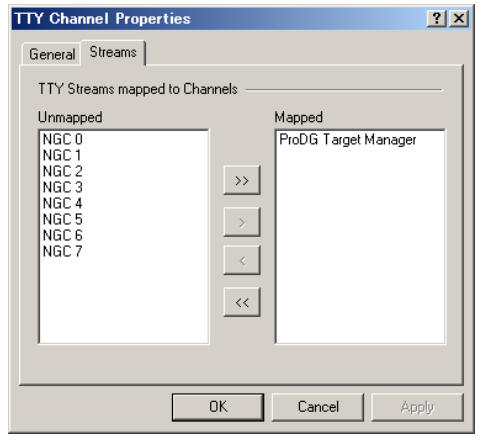
2. From the displayed context menu select **Properties...**

   The **General** tab of the Channel Properties dialog will be displayed:



Name	Enables you to rename the channel.
Visible	Toggle to determine whether the TTY channel is active. When disabled, the channel no longer appears in the left- hand tree view and TTY output is not captured for that channel.
Height	Determines the height (in lines) of the TTY channel. Increase this value to allow more output to be captured in the channel.
Width	Determines the width (in characters) of the TTY channel. This can be 80 or 132.

3. Select the **Streams** tab of the TTY Channel Properties dialog to display the following:



This page allows you to map output from several TTY streams to the currently selected channel. The left-hand list displays the currently available (unmapped) streams and the right-hand list shows those streams that are currently mapped to the selected channel.

4. The buttons between the lists enable you to move streams between them. Select as required.

   >> Moves all unmapped streams from the left to the right list.

   > Moves the currently selected unmapped stream to the right list.

   < Moves the currently selected mapped stream to the left list.

   << Moves all mapped streams from the right to the left list.

To clear a TTY channel

TTY streams are cleared automatically when you shut down Target Manager. However, the output will persist after a connection has been closed, so it may be useful to know how to clear the output from a stream when necessary.

1. Select the channel to be cleared.

2. Right-click on the right-hand pane to display the shortcut menu:

3. Click **Clear** to clear the output.

To clear all TTY channels for a target

1. Right-click the TTY directory in the left hand tree view for which you wish to clear all TTY channels.

   The following context menu will be displayed:

2. Click **Clear All** to clear all TTY channels for the selected target.

To copy TTY stream output to another application

TTY stream output can be saved to another application for later analysis using the standard Windows cut, copy and paste operations:

1. Select the channel containing the output to be copied.

2. Select the text to be copied by swiping it with the mouse button depressed, or click **Select All** or <Ctrl+A> from the TTY pane shortcut menu if you want to select everything in the pane.

3. Press <Ctrl+C> to copy the selected text to the clipboard, or click **Copy** from the TTY pane shortcut menu.

4. Press <Ctrl+V> to paste the selected text from the clipboard to the application, or select **Paste** in the application's menu.

To change the TTY channel font

The TTY channel font can be altered to suit your preference.

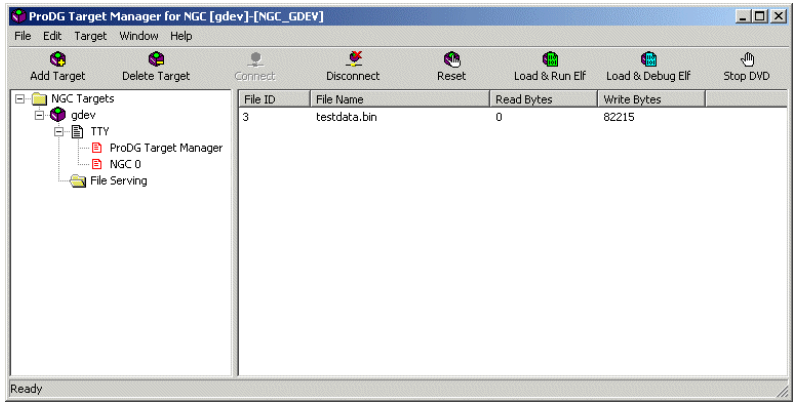
1. Select **Set Font** from the TTY pane shortcut menu.

   The Font dialog is displayed for you to set the font face, style and point size.

2. Press **OK** to accept the new settings.

The settings will apply to all TTY channels, not just the current one.

Viewing your application file serving



If you open the **File Serving** directory for the target that you are currently debugging, while your application is running on the target, you can see in real time the files that have been opened and written to, or read from, by your application. In addition the total number of bytes that have been written to each file or read from each file, while your application is running, is shown.

Setting file serving directories

File serving allows an application running on the target to retrieve files from a named directory on the host.

Target Manager provides full file serving support. The file serving directory is a property of each target and is referred to as the **File Server Root Dir.** in the Target Manager. The file serving directory for any target can be changed at any time.

When running your application on the GameCube you can view the file serving statistics in real-time in the Target Manager (see "Viewing your application file serving" on page 22).

To set the file serving directory

The file serving directory can be changed at any time.

1. Select the target for which you wish to change the file server root, in the left-hand list of GameCube targets.

2. Click **Set File Server Root** in the **Target** menu, and a folder browsing dialog appears.

3. Browse until you locate the directory that you would like to be used for file serving and click on it.

4. Click **OK**.

   The properties of the selected target should be updated in the right side of the main window to show the newly selected file server root directory under the **Current Dir** title.

It is also possible to change the file serving directory from inside your code using the following function call:

```
PCopen("SETROOT:<path>", READ_ACCESS, 0)
```

e.g.

```
PCopen("SETROOT:C:\\MyGame\\DataFiles", READ_ACCESS, 0)
```

To set the home directory

The home directory is the directory on your Win32 PC that is used to replace any file paths that you might have specified with a "~" (representing your Linux home directory) in your application source.

1. Select the target for which you wish to change the home directory, in the left hand target list in the Target Manager main window.

2. Click **Set Home Directory** from the **Target** menu, and a folder browsing dialog appears.

3. Browse until you locate the directory that you would like to be used to replace the Linux home directory (~) and click on it.

4. Click **OK**.

   The properties of the selected target should be updated in the right side of the main window to show the newly selected home directory.

It is also possible to change the home directory from inside your code using the following function call:

```
PCOpen("SETHOME:<path>", 0, 0);
```

e.g.

```
PCOpen("SETHOME:C:\\MyGame\\DataFiles", 0, 0);
```

Using a remote target

Using a remote target from the Target Manager is mostly transparent, and will work the same as a local target. However, there are a few points that you should be aware of:

- ELF files that are run or debugged from a remote target must be visible to the server. That means that the file should either be located on the server, or in a shared directory. If the file is in a shared directory you pass the full UNC path to its location in the load dialog (mapped drive letters will only work if mapped identically on the server).
- If you want to run your ELF file remotely from Visual Studio, you must first set the output filename to a location where it is visible to the server. From your project settings, select the **Link** tab and enter the full UNC path to the file in the Output file name field.
- If you want to set the DVD root when you run an ELF file, the directory you select must also be visible to the server. For DDH targets, this directory must be located on the emulation drive on the server. For NPDP-GDEV targets, the directory can be any path visible to the server.

Chapter 3: Debugging considerations

Building for debugging

Support for debugging, file serving and TTY output is provided by the libsn.a library installed by the ProDG installer. You should therefore ensure that your ELF has been built with this version of libsn.a.

For a sample program, refer to the 'SNGXDemo' example in subdirectory <install_dir>\SN Systems\ngc\SN Demos. Extract the file 'SNGXDemo.zip' to a new folder 'SNGXDemo' below 'SN Demos', then extract its contents to this new folder.

Debugging programs using DDH, NPDP-GDEV and TDEV

Starting Target Manager prior to the Debugger enables you to set up a session on the required GameCube that you can use with the Debugger. This allows you to maintain a session with a particular target even when you start and stop the Debugger.

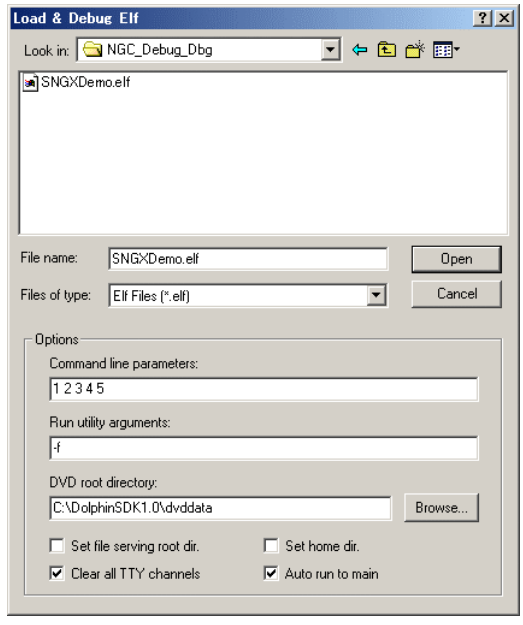
Target Manger can either be launched via the command line or Windows shortcut, or it is launched automatically by the Debugger. For any other tools (Debugger or ngcrun) to be able to interact with a particular target, it must already be configured in the Target Manager.

If you want to debug an ELF file, click the **Load & Debug ELF** button and select your ELF file from the dialog that pops up. See "Loading and debugging ELF files" on page 26 for details.

Please note that selecting the **Stop execution at main** option will stop the ELF at main() and allow you to debug from the start of your application. If this option is not selected the ELF will begin running.

Loading and debugging ELF files

You can use the Target Manager to load ELF files for later debugging using the ProDG Debugger. To do this click the **Load & Debug ELF** button on the Target Manager toolbar, or click the command in the **Target** menu. A dialog appears from which you can select the ELF file to be loaded.



Command line parameters you can enter a list of arguments to be passed to the ELF file when it is run.

Run utility arguments *(applies only to DDH and NPDP-GDEV targets)* you can pass additional switches to the command line tool that will be used to run the ELF file. For a complete list of the available switches, see the documentation for the tool you are using.

DVD root directory allows you to choose a directory to be used as the DVD emulation drive.

Set file serving root dir. sets the current directory as the file server root directory.

Set home dir. sets the current directory as the home directory.

Clear all TTY channels will clear all TTY channels for the current target before loading the ELF file.

Stop execution at main determines if the ELF should stop at main.

View the output from the **NGC0** TTY stream until the "Waiting for SN Debugger" message appears. See "Viewing TTY stream output" on page 17. You can then run the ProDG Debugger to debug the program.

Debugging an ELF file with the ProDG Debugger

Before the Target Manager can communicate with the GameCube hardware, a small section of code, known as the **debug stub**, must be downloaded and run on the target. The debug stub is linked into your ELF file from the **libsn.a** library when you build your project. In addition, you must link to the communication libraries provided by your GameCube hardware manufacturer.

- For NPDP-GDEV targets link to ode*.a.
- For DDH targets link to amc*.a.
- (For TDEV targets no debug stub is needed - just link to libsn.a as normal.)

When you debug an ELF file from the Target Manager (by clicking on the **Load & Debug ELF** button in the toolbar) the debug stub is downloaded and run on the target by the apploader. The time this takes to complete is determined by the size of your ELF file and the number of files in your dvdroot directory. For projects with large ELF files and/or many files in the dvdroot, it may be necessary to increase the load and debug timeouts from the Target Properties dialog.

Once the debug stub is executed on the target it will output the "Waiting for SN Debugger…" string to the serial TTY channel. The ELF is now ready to be debugged.

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Chapter 4: DVD emulation for ProView and SN-TDEV

Overview

ProView and SN-TDEV provide DVD emulation using Nintendo's standard DVD libraries and without any modifications to your source code. There are a few limitations, but for general-purpose DVD access this emulation should be sufficient. For testing under the real constraints of the DVD drive, you are advised to use the GDEV/DDH/NPDP-console emulators or better still the real mechanism of the NR-Reader.

The CPU overhead of DVD emulation is comparable with a real DVD drive, because interrupt driven comms are used to DMA-read the data.

The data-rate of the SN Systems custom USB2.0 link is approximately 3MB/s, which approximates to the maximum data-rate of a real DVD drive (inner edge: 2MB/s; outer edge: 3MB/s).

The following file types are automatically aligned on the disc to 32KB:

.adp
.adpcm
.dat
.hvqm
.pcm
.wav

All other file types are aligned to 32 bytes.

When using ProView you can load assets (data files, models etc.), via the DVD functions instead of using fileserving commands.

Limitations

SN Systems' DVD emulation has the following limitations:

- No multi-disc support.
- No support for the hardware streamed ADPCM by the DVD mechanism - DVDxxxStream() functions. These functions are ignored.
- No seek time or data rate emulation – data transfers occur at the maximum speed of the USB2.0 link.
- No support for layout of files on the disc. Files and subdirectories placed in the 'DvdRoot' directory are sorted into alphabetical order before being built into the final DVD image.
- DVD emulation uses hardware breakpoints, so you can't use these for debugging during emulation.

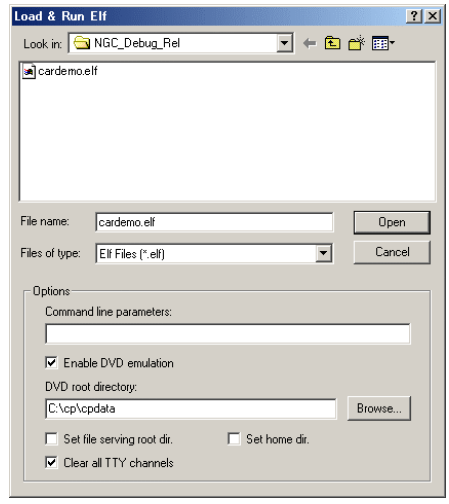
Building your ELF

1. Build your ELF with libsn.a that supports DVD emulation, version 45 or later.
2. Place all your DVD data files into a dedicated 'DvdRoot' directory on your PC's HDD. This can consist of nested subdirectories, mimicking the directory tree that you want on your emulated disc image. The number of files and directories is only limited by the size of your FST. See the Nintendo documentation for details about calculating the size of the FST.
3. The function snIsSNTDEV() allows you to determine if your ELF has been run by SN-TDEV, e.g:

```
if(snIsSNTDEV())
   OSReport("Running on SN-TDEV\n");
```

Loading your ELF

1. If using ProView, download your ELF using the Target Manager. If using SN-TDEV, download your ELF using the Target Manager, Debugger or Visual Studio Integration.
2. Enable DVD Emulation and set the DvdRoot in the Download dialog. Any Nintendo DVD functions that you now call will get serviced by the host PC over the USB2.0 link.



3. When the **Enable DVD emulation** button is not checked, the **DVD root directory** selection is disabled and any DVD operations executed on the target will go through the default mechanism (e.g. the real DVD drive on the TDEV).

4. You can view the file currently open for reading on the DVD image by selecting the File Serving view in the Target Manager user interface.

5. You can terminate DVD emulation from the Target Manager by pressing the **Stop DVD** button. To re-enable it, you will have to re-download your ELF.

6. During emulation, the currently open file from the DVD image is locked for write access by the Target Manager to prevent the file changing at run-time. After a DVD session you should select the **Stop DVD** button (or run "ngctm –t mytdev –v" from the command line) to ensure that all locked handles are released.

Target Manager supports a new command line option, -e, to enable DVD emulation when running an ELF from the command prompt.

Examples:

```
ngctm –t mytdev –x mygame.elf –d "C:\mygame\dvdroot" –e
```

loads and runs mygame.elf on the mytdev target with DVD emulation enabled from C:\mygame\dvdroot.

```
ngctm –t mytdev –xr mygame.elf –d "C:\mygame\dvdroot" –e
```

loads and runs mygame.elf (in debug mode) on the mytdev target with DVD emulation enabled from C:\mygame\dvdroot.

Supported DVD functions

The following table lists the DVD functions supported by SN Systems' emulation and compares it with that supported by Nintendo's DVDETH library (9-1-2003) using the Broadband Adapter.

DVD Function	Nintendo supported?	SN Systems supported?
DVDInit	√	√
DVDOpen	√	√
DVDRead	√	√
DVDReadAsync	√	√
DVDConvertPathToEntrynum	X	√
DVDFastOpen	X	√
DVDSeek(Async)	√	√
DVDClose	√	√
DVDGetLength	√	√
DVDGetStartAddr	√	√
DVDGetTransferredSize	X	√
DVDSetAutoInvalidation	√	√
DVDChangeDir	X	√
DVDGetFSTLocation	X	√
DVDRead(Async)Prio	√	√
DVDSeek(Async)Prio	√	√
DVDOpenDir	X	√
DVDFastOpenDir	X	√
DVDReadDir	X	√
DVDCloseDir	X	√
DVDTellDir	X	√
DVDSeekDir	X	√
DVDRewindDir	X	√
DVDPrepareStream(Async)	X	X
DVDCancelStream(Async)	X	X
DVDStopStreamAtEnd(Async)	X	X
DVDGetStreamErrorStatus(Async)	X	X

DVDGetStreamPlayAddr(Async)	X	X
DVDCancel(Async)	X	√
DVDCancelAll(Async)	X	√
DVDPause	X	√
DVDResume	X	√
DVDDumpWaitingQueue	√	√
DVDGetUserData	√	√
DVDSetUserData	√	√
DVDGetCurrentDiskID	X	√
DVDGenerateDiskID	X	√
DVDCompareDiskID	X	√
DVDChangeDisk	X	X
DVDGetFileInfoStatus	√	√
DVDGetCommandBlockStatus	√	√
DVDGetDriveStatus	X	√
DVDSetAutoFatalMessaging	X	√
DVDCheckDisk	X	√
Total DVD functions supported	16	**38**

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Chapter 5: The ngcrun command-line utility

Overview

The ngcrun program is a command-line tool for communicating with a GameCube target. It is used in conjunction with the Target Manager application (ngctm.exe), which must be running for ngcrun to work.

The ngcrun program has options which can reset the target, and load and debug an ELF file.

ngcrun command-line syntax

ngcrun can either be used directly from the command line, or in batch or make files.

The ngcrun command line is as follows:

```
ngcrun <options> [<file> [<args>]]
```

Use <file> [and optional <args>] to specify the name of the executable ELF file (and optional arguments to that program) which you wish to run on the target.

Omitting <file> and <args> just performs the specified options.

The following options may be entered as arguments to ngcrun:

-p	Enables you to specify that stdout stream is displayed. No ELF file is necessary with this switch.
-q	Enables you to specify that ngcrun is in quiet mode providing there are no errors; ngcrun prints a progress report unless the -q option is specified.
-r	Resets the target.
-t <name>	Enables you to specify the target to connect to using its name (as shown in the Target Manager, see "To add a new target" on page 8). The name must be enclosed in quotes if it has spaces in it (e.g. "my npdp-gdev").

	If you enter a name that cannot be identified in the list of targets on the Target Manager, or you do not enter a name, then a dialog appears asking you to select from the available targets when you launch the Debugger. Note that if the name of the target contains spaces you must enclose it in double quotes on the command line.
-x	Enables you to specify that the code is executed on the target after load. This is the default behavior when you load an ELF file.
-xd	Load and debug ELF file.

The return code of the program is 0 if everything worked or 1 if there were any errors.

Showing TTY output

Debugging with ngcrun is most easily achieved using two instances of the program running in separate command prompt sessions.

In one command prompt window use the command:

```
ngcrun -t <name> -p
```

to intercept all TTY output from your target.

From a second command prompt window you can then reset the target and run the program you wish to debug by invoking the command:

```
ngcrun -t <name> -x <file>
```

where <file> is the name of the ELF file to be executed.

Appendix A1: TDEV console

About the TDEV console

This section contains information which is specific to the TDEV console.

The TDEV is a cost-effective hardware platform for developing games for the GameCube. The TDEV console closely resembles a retail GameCube, but has the added advantage of an extra 24 MB of memory (48 MB in total) and a slot for our USB2.0 adapter for loading games (so there is no need to burn discs or cartridges).

The TDEV console can be used with these SN Systems applications:

- ProDG
- Tuner

ProDG is used to compile your game code, then execute and debug it on the TDEV. Tuner can be used to fine-tune your game while it is running on the TDEV console.

TDEV console

Features

SN-TDEV is supplied with host and target software. The host software, Target Manager, provides facilities for loading and running ELFs. The target software (supplied on an NR disc) is used to boot the console and control communications with the host software.

- Debugging support
- Download and run game code
- Full fileserving support
- TTY output (with color support)
- SDK and command-line facilities
- Fast USB2.0 connection (see below)
- Extra memory (see below)

Our custom USB2.0 connection to the TDEV console provides fast data transfer, with peak read and write speeds of 3 MB per second. When using our custom USB2.0 connection, the BBA (Broadband Adapter) socket is left free, allowing you to develop online games. Our USB2.0 connection is also compatible with many off-the-shelf USB2.0 adapters, including the Adaptec® USB2connect™ PCMCIA adapter, model number AUA-1420, so you can use SN-TDEV with a laptop for demonstrations and exhibitions.

The TDEV console has 48 MB of RAM, allowing more room for your debugging and testing needs. NR-Readers, NPDP consoles and retail GameCubes have 24 MB of RAM.

Installation

For installation instructions, see the separate *Installation Guide for Nintendo GameCube Consoles*.

Parts list

SN-TDEV consists of:

- TDEV console and controller
- Host and target software
- USB2.0 adapter and cable (custom)
- USB2.0 PCI card

System requirements

- Windows NT4 SP6, Windows 2000 or Windows XP
- Nintendo GameCube Software Development Kit (SDK)

File serving with SN-TDEV

See "Setting file serving directories" on page 22 for information on how to set the file serving and home directories.

SN-TDEV provides file serving support through the SN file serving functions PCcreat, PCopen, PCclose, PCread, PCwrite and PClseek functions. Your program must include the libsn.h header file and link to libsn.a to use these functions, e.g.:

```
int fd;
char buff[0x1000]; // Buffer for reading into

fd = PCOpen("test.txt", 0);
if (fd >= 0)
{
  // Read a small amount
  int ret = PCRead(fd, buff, 100);
  if (ret > 0)
    PCwrite(-1, "Read succeeded\n");
  else
    PCwrite(-1, "Read failed\n");

  PCClose(fd);
}
```

For a sample program, refer to the 'fs_test' example in subdirectory <install_dir>\SN Demos\ .. \fs_test.

TTY support

See "Viewing TTY stream output" on page 17 for information on how Target Manager handles TTY output from the console.

The TDEV boot program also patches the Nintendo OSReport function and the printf function in libc.a, so you can also use these functions to output TTY.

Creating a TDEV boot disc

This section describes the process necessary to create boot discs for the TDEV console. Also, you may want to burn your own data files onto a boot disc so you can access them using the Nintendo DVD functions.

The files required to create a console boot disc for your territory are installed to <install_dir>\boot. In addition, you will need the following tools:

- TickleBI2 (from Nintendo GameCube SDK)
- OdemMakeDlf.exe (from NPDP-GDEV install)
- ddpack.exe (from Nintendo's website)
- MakeGCM.exe (from Nintendo's website)
- NR-Writer software (from Nintendo's website)

Now follow these instructions:

1. Configure the DVD root definition in sndvd.ddf, for example:

   ```
   [Input]
   DvdRoot="C:\cp\cpdata\cardemo"
   ```

2. Configure SN-TDEV for your territory.

   MakeSNDVD <jp|us> [gcm]

   where jp = Japan, and us = USA.

   This will create sndvd.dpf and optionally the 1.4GB sndvd.gcm file. You will need ddpackD and makegcm in the path.

3. Finally burn the image with the NR-Writer software.

Appendix A2: NPDP console

About the NPDP console

This section contains information which is specific to the NPDP console.

The NPDP console is supplied by Nintendo. See **www.warioworld.com** for details.

The NPDP console can be used with the following SN Systems application:

- ProView

ProView is used to enable fast downloads and previewing of game artwork on the console, without the time and expense of burning the game to disc.

Features

The host software, Target Manager, provides facilities for loading and running ELFs. The target software (supplied on an NR disc) is used to boot the console and control communications with the host software.

- download and run game code
- full fileserving support
- TTY output (with color support)
- SDK and command-line facilities
- fast USB2.0 connection (see below)

Our custom USB2.0 connection to the NPDP console provides fast data transfer, with peak read and write speeds of 3 MB per second. When using our custom USB2.0 connection, the BBA (Broadband Adapter) socket is left free. Our USB2.0 connection is also compatible with many off-the-shelf USB2.0 adapters, including the Adaptec® USB2connect™ PCMCIA adapter, model number AUA-1420, so you can use ProView with a laptop for demonstrations and exhibitions.

Installation

For installation instructions, see the separate *Installation Guide for Nintendo GameCube Consoles*.

System requirements

- Windows NT4 SP6, Windows 2000 or Windows XP
- Nintendo GameCube Software Development Kit (SDK)

Creating an NPDP boot image

This section describes the process necessary to create a boot image for the NPDP console. Also, you may want to burn your own data files onto a boot disc so you can access them using the Nintendo DVD functions.

The files required to create a boot image for your territory are installed to <install_dir>\boot. In addition, you will need the following tools:

- TickleBI2 (from Nintendo GameCube SDK)
- OdemMakeDlf.exe (from NPDP-GDEV install)
- ddpackD.exe (from Nintendo's website)
- MakeGCM.exe (from Nintendo's website)
- NPDP-Writer software (from Nintendo's website)

Now follow these instructions:

1. Configure the DVD root definition in sndvd.ddf, for example:

   ```
   [Input]
   DvdRoot="C:\cp\cpdata\cardemo"
   ```

2. Configure NPDP for your territory.

   MakeSNDVD <jp|us> [gcm]

   where jp = Japan, and us = USA.

   This will create sndvd.dpf and optionally the 1.4GB sndvd.gcm file. You will need ddpackD and makegcm in the path.

3. Finally burn the image with the NPDP-Writer software.

Appendix B: Reference

Keyboard shortcuts

Many of the most frequently used commands in the Target Manager can be accessed via keyboard shortcuts. Currently the keyboard shortcuts are not customizable.

Keyboard shortcut	Description	Context
c	Connects to the currently selected target.	A target must be selected in the target manager that you are not currently connected to.
d	Disconnect from the currently selected target.	You must be connected to the currently selected target.
e	Load and run ELF.	You must be connected to the currently selected target.
r	Resets the currently selected target.	You must be connected to the currently selected target.
x	Load and run IRX.	You must be connected to the currently selected target.
Insert	Add a new target.	Any
Delete	Removes the currently selected target.	The target to be deleted must be selected.
Tab/Shift+Tab	Toggles focus between the left and right parts of the target manager main pane.	Any
F5	Refresh connection status.	Any

Configuration file: NGCTM.INI

The Target Manager INI file (ngctm.ini) contains the current Target Manager program and installed target settings. It is not usually necessary to edit the ngctm.ini file directly.

Setting level of debug logging

You can set the level of debug logging by changing the following parameter:

"Logging" Determines the level of debug logging output by the Target Manager. The value is interpreted as a decimal bit field:

Bit Flag (Decimal)	*Debugging level*
1	Log all communications with targets
2	Log all host PC commands
4	Log file serving activity
8	Output log to file rather than TTY
16	Log update notifications

For example, the value 13 (1 + 4 + 8) would log all target comms and file serving activity to a log file.

SDK support

Target Manager SDK allows you to develop applications that communicate to GameCube development hardware via the Target Manager. It provides a simple to use API, available in both a static and dynamically linked library.

The SDK provides functions to:

- Enumerate available targets
- Connect to/Disconnect from a target
- Download and run ELF files on the target
- Stop and start execution of the target
- Get and set target registers
- Get and set target memory

- Capture TTY output from the target
- Receive target status information

The SDK header file, ngctmapi.h, contains a full listing of the functions available and a detailed description of their operation.

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